

Real-time Continuous Perceived Difficulty Annotation during Video Learning Using a More Natural Interface

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Abstract: In video-based learning, students' perceived difficulty often varies over time. Capturing these dynamics enhances learning and instructional design. Existing continuous annotation tools often require devices that are not commonly available and may add unnecessary cognitive load. To overcome these challenges, we developed a more natural annotation interface that leverages mouse scroll and a shadow effect. A subsequent user study demonstrated its feasibility in educational practice. Lastly, we outlined its potential applications for learners and instructors.

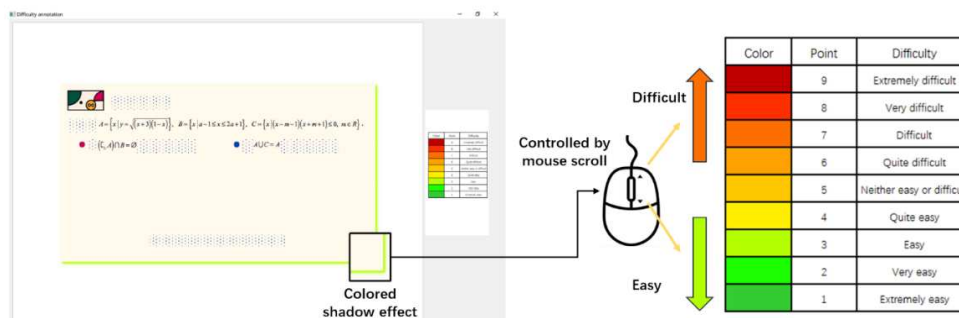
Introduction

Video has emerged as an essential and popular tool in education, enriching blended learning environments and engaging digital-native learners. Since the advent of video-based learning, its effectiveness has been debated, particularly concerning the insufficient provision of feedback. Although post-video tests and surveys offer valuable insights, they inadequately capture the learning dynamics. Notably, within these dynamics, the perceived difficulty may fluctuate over time, as suggested by the theories on instantaneous cognitive load (Xie & Salvendy, 2000). Analyzing dynamic patterns of perceived difficulty can enhance knowledge consolidation for learners, while recognizing common patterns can guide teachers in instructional design.

Recent studies have utilized real-time continuous annotation tools to record psychological states during video watching. For instance, Girard and C. Wright (2018) introduced an annotation tool employing a joystick to capture emotions in real-time. Similarly, Srivastava et al. (2019) created a tool for instant difficulty rating of video lectures using a physical slider. These innovations have enhanced the collection of dynamic user responses throughout the learning process. However, the limited availability of joysticks and sliders restricts their utility in educational contexts. Moreover, the separate visual interfaces for annotation and video watching in these tools could induce extraneous cognitive load, potentially impacting the learning experience. Therefore, we introduced a more natural interface using mouse scroll and shadow effects to overcome these limitations and enhance the learning experience. A pilot user study was subsequently carried out to examine the feasibility of the tool.

Methods

Figure 1
Design and Implementation of the Annotation Tool



In our system design (Figure 1), the mouse is utilized to annotate the difficulty level through vertical scrolling. The annotations correspond to the perceived difficulty scale (Sweller et al., 2011), which uses a 9-point Likert scale ranging from 1 (extremely easy) to 9 (extremely difficult). To minimize the visual separation between annotation and video viewing, we implement a colored shadow effect to represent the annotated difficulty levels. A shift towards green denotes easier content, suggesting relaxation, whereas a transition to red indicates more challenging material, correlating with stress. This user-friendly interface enables learners to intuitively annotate their perceived difficulty levels.

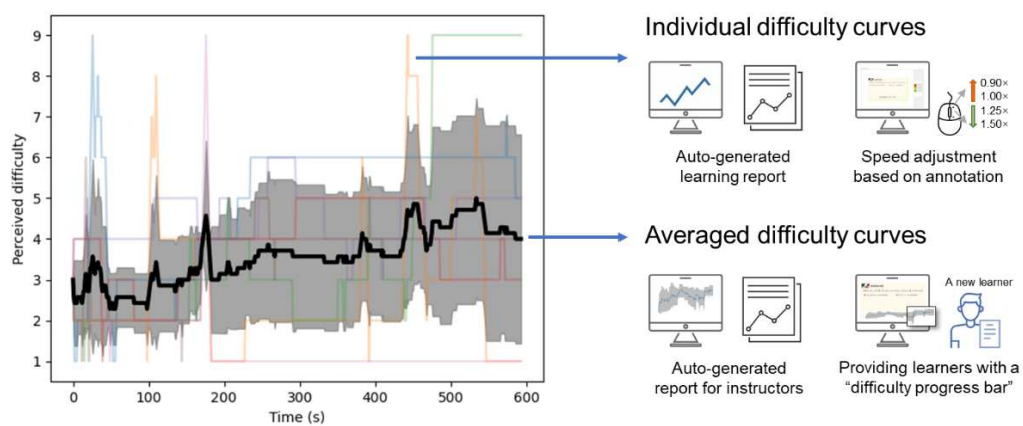
In the subsequent pilot user study, seven students from the same blended learning course were recruited. They annotated three video clips which were selected from the course material, and then filled out a questionnaire to assess the overall difficulty with the same scale. To analyze the dynamics of the annotations, the mean and quartile values of the curves were calculated, and they were then compared to the overall difficulty ratings.

Results and discussion

The average value of the curve showed a strong correlation with the overall ratings (Pearson's $r = 0.815$, $p < 0.001$, $n = 21$), suggesting the tool's feasibility in capturing instantaneous perceived difficulty. Notably, this correlation coefficient exceeded those using quartile values (median: $r = 0.764$, lower quartile: $r = 0.751$, upper quartile: $r = 0.762$). These results are consistent with the theoretical relationship between instantaneous and overall cognitive load (Xie & Salvendy, 2000).

Figure 2

A Sample Difficulty Curve and Its Potential Applications



Moving beyond the statistical analysis, we further explored the data through the visualization of individual and averaged difficulty annotation curve for the same video, as illustrated by the example in Figure 2. In the beginning of the video, three distinct difficulty peaks were observed, indicating that these students encountered challenges with different concepts. Analysis of the averaged curve revealed an increasing trend in difficulty, which is consistent with the instructor's expectation.

Drawing on the results of the pilot study, several potential applications are proposed based on the annotation tool, as shown in Figure 2. In a prospective application, individual learning curves could inform an intelligent tutoring system to automatically generate a learning report, identifying challenging concepts for the learner and providing targeted explanations. Additionally, the system might be envisioned to dynamically adjust the video speed in response to annotations of difficulty, thereby creating a suitable pace for learners. Similarly, the averaged curve could offer valuable insight into collective learning challenges, which could potentially tell teachers which concepts are difficult to comprehend, as well as providing a "difficulty progress bar" to assist new learners in navigating challenging parts of a video lesson.

References

- Girard, J. M., & C. Wright, A. G. (2018). DARMA: Software for dual axis rating and media annotation. *Behavior research methods*, 50, 902-909.
- Srivastava, N., Velloso, E., Lodge, J. M., Erfani, S., & Bailey, J. (2019). Continuous evaluation of video lectures from real-time difficulty self-report. *Proceedings of the 2019 CHI conference on Human factors in computing systems*,
- Sweller, J., Ayres, P., Kalyuga, S., Sweller, J., Ayres, P., & Kalyuga, S. (2011). Measuring cognitive load. *Cognitive load theory*, 71-85.
- Xie, B., & Salvendy, G. (2000). Review and reappraisal of modelling and predicting mental workload in single- and multi-task environments. *Work & stress*, 14(1), 74-99.

Acknowledgments

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